TASFIA JANNAT TASFI

LLM Reasoning | Natural Language Processing | Human-Centered AI

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Research Profile

Computer Science and Engineering graduate with research experience in multilingual LLM reasoning, NLP evaluation, and human-AI decision-making. Co-developed a Bengali-English benchmark for creative reasoning and studied how explanations affect people's judgments of AI advice. Interested in why language models struggle to generalize across problem formulations and how human reasoning can inform better AI systems.

Research Interests: LLM Reasoning; Systematic and Compositional Generalization; Multilingual NLP; Human-Machine Intelligence; Human-AI Decision-Making; Behavioral Evaluation

Education

B.Sc. in Computer Science and Engineering **11 Aug. 2026**
Rajshahi University of Engineering & Technology (RUET) *Rajshahi, Bangladesh*
CGPA: 3.77/4.00 | **Program:** BAETE-accredited CSE program

Research Experience

Multilingual Creative Reasoning and Lateral Thinking in Instruction-Tuned LLMs **2026**
Language Model Evaluation | Bengali-English Reasoning | Error Analysis

- Co-developed a parallel Bengali-English benchmark to study how instruction-tuned language models respond to creative reasoning and lateral-thinking tasks.
- Analyzed model failures involving mistranslation, cultural mismatch, and logical inconsistency across languages and task presentations.

Undergraduate Thesis: Designing and Evaluating Explanation Interfaces for AI-Assisted Medical Misinformation Detection
NLP | Controlled User Study | Explainable AI | Human-AI Interaction

- Developed a RoBERTa-based detector using **17,611** multi-source examples, probability calibration, and conformal selective prediction; handled **91.56%** of cases automatically with **0.9806 macro-F1** on the accepted subset.
- Designed four interactive explanation interfaces showing predictions, linguistic cues, uncertainty, and supporting evidence; implemented the research prototype with **Next.js and FastAPI**.
- Evaluated the interfaces in a **60-participant, 1,200-trial** controlled study. The most informative condition increased decision accuracy from **73.7% to 90.0%** and reduced reliance on incorrect AI advice from **77.8% to 37.8%**.

Publications

Nahid Montasir Rifat, Tasfia Jannat Tasfi, Foyez Ahmed Dewan, and Barshon Sen. *Evaluating Creative Reasoning and Lateral Thinking in Instruction-Tuned LLMs for Bengali.* 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), 2026.
DOI: https://doi.org/10.1109/QPAIN69676.2026.11546604

Co-developed a parallel Bengali-English benchmark to examine creative reasoning and lateral-thinking tasks in instruction-tuned language models. Analyzed cross-language failures involving translation, cultural context, and logical consistency, highlighting challenges in evaluating generalizable reasoning.

Jabunnesa Jahan Sara, Shahriar Rizvi, and Tasfia Jannat Tasfi. *Beyond Artifacts: Semantic Deepfake Detection and Digital Provenance Tracking with Multimodal Large Language Models.* 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025.
DOI: https://doi.org/10.1109/ICCIT68739.2025.11490141

Contributed to a multimodal approach for identifying manipulated digital media through semantic analysis and web-based provenance checks. The study explored how language and visual evidence can support credibility assessments beyond image artifacts alone.

Selected Achievements

International Technology Competition: Huawei Seeds for the Future Regional Round, China (2024)
Participated in the international regional competition with a team proposal for IoT-enabled water-quality monitoring and purification. The design combined sensor-based monitoring (pH, turbidity, conductivity, dissolved oxygen, and temperature), microcontroller-connected hardware, GSM/cloud data transfer, and automated motor/valve control.
Press: The Daily Star: Bangladeshi team at the China regional round (thedailystar.net)

- **Champion, Huawei Seeds for the Future Tech-Business Competition, Bangladesh, 2024.**
- **1st Place, UIHP 5th Cohort (WayMakers)**: startup innovation and MVP development.
- **Champion, Hult Prize at RUET, 2023.**
- **Government General Scholarship**: all four undergraduate years.
- **Finalist, International Youth Math Challenge**: top 20% globally, 2024.
- **70% Merit Scholarship, ISCEA PTAK Prize, 2024.**
- **Government Talent Pool Scholarship**: secondary-school academic distinction.

Selected Systems & Design Projects

BongoSeek: LLM-Powered Conversational Search. Built a conversational search prototype using context-aware queries and dialogue memory for multi-turn interaction. **Code:** github.com/Shion115/Bongo_Seek

EarlyMind: AI-Assisted Child Development Interface. Created adaptive dashboards and progress feedback for an interactive child-development assessment interface. **Code:** github.com/Shion115/EarlyMind

Professional & Leadership Experience

Co-Founder, Delibro **Nov. 2024 to Present**
Community-Based Delivery Platform

- Co-founded a peer-to-peer delivery platform and contributed to interface design, prototyping, and user-oriented validation around trust and coordination; supported by the World Bank-supported University Innovation Hub.

Official Ambassador **Jun. 2025 to Present**
International Computer Science Competition

- Organized programming outreach, mentorship, and student initiatives. **Profile:** icscompetition.org/en/ttasfi

Industrial Trainee **Mar. 2025**
BRAC IT

- Worked on frontend interface design and web-development workflows.

App Developer **Jun. 2022 to Jan. 2023**
bdapps

- Developed and deployed PHP applications integrating frontend and backend functionality.

Research & Technical Skills

NLP & Modeling	Python, RoBERTa, Transformers, Hugging Face, scikit-learn, LLM Evaluation, Cross-Lingual Error Analysis
Research Methods	Controlled Experiments, Behavioral Evaluation, User Studies, Decision Accuracy, Trust/AI Reliance, Calibration, Conformal Prediction
Development	Next.js, FastAPI, Flask, JavaScript, HTML/CSS, Figma, C++, PHP, Git, Jupyter Notebook, Google Colab

References

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